

DS Lab-01

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Task 1:

```
#include<iostream>
using namespace std;
void removeDuplicates(int*, int&);
void countFrequency(int*, int, int[][100]);
void descendingFrequency(int*, int, int[][100]);
void printArray(int*, int, int[][100]);

int main(){
    cout<<"24P-3017"<<endl;

    int array[]={4, 5, 6, 5, 4, 3, 4, 6, 7, 3, 5};
    int size=sizeof(array)/sizeof(array[0]);
    int frequency[2][100]={0};

    countFrequency(array, size, frequency);
    removeDuplicates(array, size);
    descendingFrequency(array, size, frequency);
    cout<<"Rearranged Array: ";
    printArray(array, size, frequency);

}

void removeDuplicates(int *array, int &size){
    for(int i=0;i<size;i++){
        for(int j=0;j<size;j++){
            if(j==i) continue;
            if(array[i]==array[j]){
                while(j<size){
                    array[j]=array[j+1];
                    j++;
                }
                size--;
            }
            else continue;
        }
    }
}
```

```
}
```

```
void countFrequency(int *array, int size, int frequency[][100]){  
    int n=0;
```

```
    for(int i=0;i<size;i++){  
        bool exists=false;  
        for(int k=0;k<size;k++){  
            if(frequency[0][k]==array[i]){  
                exists=true; break;  
            }  
            else continue;  
        }  
        if(exists==true) continue;
```

```
        frequency[0][n]=array[i];
```

```
        int counter=0;  
        for(int j=0;j<size;j++){  
            if(array[j]==array[i]) counter++;  
            else continue;  
        }  
        frequency[1][n]=counter;  
        n++;
```

```
    }  
}
```

```
void descendingFrequency(int *array, int size, int frequency[][100]){
```

```
    for(int i=0;i<size;i++){  
        if(frequency[0][i]==0 && frequency[1][i]==0) break;  
        for(int j=i+1;j<size;j++){  
            if(frequency[1][j]>frequency[1][i]){  
                swap(frequency[0][j], frequency[0][i]);  
                swap(frequency[1][j], frequency[1][i]);  
            }  
            else if(frequency[1][j]==frequency[1][i]){  
                if(frequency[0][j]<frequency[0][i]) swap(frequency[0][j], frequency[0][i]);  
                else continue;  
            }  
        }  
    }
```

```
    }  
}
```

```
for(int i=0;i<size;i++){  
    if(frequency[0][i]==0 && frequency[1][i]==0) break;
```

```

        array[i]=frequency[0][i];
    }
}

void printArray(int *array, int size, int frequency[][100]){
    cout<<"[";
    for(int i=0;i<size;i++){
        if(i==size-1) cout<<array[i];
        else cout<<array[i]<<" ";
    }
    cout<<"]\n\n";

    cout<<"Frequencies:\n\n";
    for(int i=0;i<size;i++){
        if(frequency[0][i]==0 && frequency[1][i]==0) break;
        cout<<frequency[0][i]<<" -> "<<frequency[1][i]<<endl;
    }
}

```

Output:

```

24P-3017
Rearranged Array: [4, 5, 3, 6, 7]

Frequencies:

4 -> 3
5 -> 3
3 -> 2
6 -> 2
7 -> 1

```

Task 2:

```
#include<iostream>
using namespace std;
void reverseSentence(char*, int);
void reverseWord(char*, int);
void convertCase(char*, int);
void removeSpaces(char*, int);
void printArray(char*, int);

int main(){
    cout<<"24P-3017"<<endl;

    char string[]={'H', 'e', 'l', 'l', 'o', ' ', 'W', 'o', 'r', 'l', 'd', '\0'};
    int length=sizeof(string)/sizeof(string[0]);

    reverseSentence(string, length);
    removeSpaces(string, length);
    convertCase(string, length);
    printArray(string, length);
}

void reverseSentence(char *string, int length){
    for(int i=0;i<length/2;i++){
        swap(string[i], string[length-2-i]);
    }
}

void reverseWord(char *string, int length){
    int firstLetter=0;
    int lastLetter;

    for(int i=0;i<length;i++){
        int n=0;
        if((string[i]==' ') || (string[i+1]=='\0')){
            lastLetter=i-1;
            for(int j=firstLetter;j<lastLetter/2;j++){
                swap(string[firstLetter+n], string[lastLetter-n]);
                n++;
            }
            firstLetter=i+1;
        }
    }
}
```

```
}
```

```
void convertCase(char *string, int length){
```

```
    for(int i=0;i<length;i++){
```

```
        if(string[i]==' ' || string[i]=='\0') continue;
```

```
        if(string[i]<96) string[i]+=32;
```

```
    }
```

```
    for(int i=0;i<length;i++){
```

```
        if(string[i]=='a' || string[i]=='e' || string[i]=='i' || string[i]=='o' || string[i]=='u' ) string[i]-=32;
```

```
    }
```

```
}
```

```
void removeSpaces(char *string, int length){
```

```
    bool space=true;
```

```
    do{
```

```
        for(int i=0;i<length;i++){
```

```
            if(string[i]==' '){
```

```
                while(i<length){
```

```
                    string[i]=string[i+1];
```

```
                    i++;
```

```
                }
```

```
                space=false;
```

```
            }
```

```
            else continue;
```

```
        }
```

```
        for(int i=0;i<length;i++){
```

```
            if(string[i]==' '){
```

```
                if((string[i+1]=='\0') || (string[i+1]==' ')) break;
```

```
                else space=true;
```

```
            }
```

```
            else continue;
```

```
        }
```

```
    }while(space==true);
```

```
}
```

```
void printArray(char *string, int length){
```

```
    for(int i=0;i<length;i++){
```

```
        if(string[i]=='\0') break;
```

```
        else cout<<string[i];
```

```
    }
```

```
}
```

Output:

```
24P-3017
dlrOwOllEh
```

note: according to the question this is the correct output (vowels to uppercase and consonants to lowercase)

Task 3:

```
#include<iostream>
using namespace std;

int main(){
    int rollNo=3017;

    int *teacher=&rollNo;
    int **incharge=&teacher;
    int ***principal=&incharge;

    cout<<"24P-3017"<<endl;
    cout<<"\nDirect roll number: "<<rollNo;
    cout<<"\nRoll number through teacher: "<<*teacher;
    cout<<"\nRoll number through class incharge: "<<**incharge;
    cout<<"\nRoll number through principal: "<<***principal;
}
```

Output:

```
24P-3017
Direct roll number: 3017
Roll number through teacher: 3017
Roll number through class incharge: 3017
Roll number through principal: 3017
```

Task 4:

```
#include<iostream>
using namespace std;
int findSecondLargest(int*, int);
void countEvenOdd(int*, int, int*, int*);
void rotateArray(int*, int, int);
int sumOfPrimes(int*, int);
void printArray(int*, int);

int main(){
    cout<<"24P-3017\n\n";

    int size;
    cout<<"Enter size: "; cin>>size;
    int array[size];
    cout<<"Enter elements: ";

    for(int i=0;i<size;i++){
        cin>>array[i];
    }

    int *evenCount=new int;
    int *oddCount=new int;
    *evenCount=0;
    *oddCount=0;

    printArray(array, size);
    cout<<"Second Largest Element: "<<findSecondLargest(array, size)<<endl;
    countEvenOdd(array, size, evenCount, oddCount);
    cout<<"Even Numbers: "<<*evenCount; cout<<"\nOdd Numbers: "<<*oddCount;
    cout<<"\nSum of Prime Numbers: "<<sumOfPrimes(array, size);
    rotateArray(array, size, 2);
    cout<<"\nArray rotated by 2 positions to the left: ";
    printArray(array, size);

    delete evenCount;
    evenCount=NULL;
    delete oddCount;
    oddCount=NULL;
}

int findSecondLargest(int *array, int n){
```

```

int secondLargest=-1;

for(int i=0;i<n;i++){
    for(int j=0;j<n-1-i;j++){
        if(*(array+j)>*(array+j+1)) swap(*(array+j), *(array+j+1));
        else continue;
    }
}

int largest=array[n-1];
for(int i=n-2;i>0;i--){
    if(array[i]<largest){
        secondLargest=array[i];
        break;
    }
}

return secondLargest;
}

void countEvenOdd(int *array, int n, int *evenCount, int *oddCount){
    for(int i=0;i<n;i++){
        if(*(array+i)%2==0) *evenCount+=1;
        else *oddCount+=1;
    }
}

void rotateArray(int *array, int n, int k){
    int temp[k];

    for(int i=0;i<k;i++){
        *(temp+i)=*(array+i);
    }
    for(int i = 0; i < n - k; i++){
        *(array+i)=*(array+i+k);
    }
    for(int i=0;i<k;i++){
        *(array+n-k+i)=*(temp+i);
    }
}

int sumOfPrimes(int *array, int n){
    int sum=0;

```



```

for(int i=0;i<n;i++){
    if(*(array+i)<=1) continue;

    else if(*(array+i)==2){
        sum+=*(array+i);
        continue;
    }

    else if(*(array+i)%2==0) continue;

    bool isPrime=true;
    for(int j=3;j*j<=*(array+i);j+=2) {
        if (*(array+i)%j==0){
            isPrime=false;
            break;
        }
    }

    if(isPrime==true) sum+=*(array+i);
}

return sum;
}

void printArray(int *array, int n){
    cout<<"[";
    for(int i=0;i<n;i++){
        if(i==n-1) cout<<*(array+i);
        else cout<<*(array+i)<<" ";
    }
    cout<<"]\n\n";
}

```

Output:

```

24P-3017

Enter size: 5
Enter elements: 7 2 14 6 9
[7, 2, 14, 6, 9]

Second Largest Element: 9
Even Numbers: 3
Odd Numbers: 2
Sum of Prime Numbers: 9
Array rotated by 2 positions to the left: [7, 9, 14, 2, 6]

```